

## A Catch in the Reye

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**ABSTRACT.** Twenty-six cases of Reye syndrome from The Children's Hospital, Camperdown, Australia, occurring between 1973 and 1982 were reviewed. Of these, 20 cases met the US Public Health Service Centers for Disease Control criteria for the diagnosis of Reye syndrome. Aspirin or salicylate ingestion had occurred in only one of the 20 cases (5%), and paracetamol (acetaminophen) had been administered in only six of the cases (30%). Pathologic confirmation of the diagnosis of Reye syndrome was accomplished in 90% of the cases. The incidence of Reye syndrome in New South Wales, Australia, is estimated from this study to be approximately nine cases per 1 million children compared with recent US data of ten to 20 cases per 1 million children and three to seven cases per 1 million children in Great Britain. The mortality for these Reye syndrome cases in Australia was 45% as compared with a 32% case-fatality rate in the United States. In Australia, the pediatric usage of aspirin has been extremely low for the past 25 years (less than 1% of total dosage units sold), with paracetamol (acetaminophen) dominating the pediatric analgesic and antipyretic market. Reye syndrome may be disappearing from Australia despite a total lack of association with salicylates or aspirin ingestion, since there were no cases found at The Children's Hospital in 1983, 1984, or 1985. *Pediatrics* 1987;80:638-642; *Reye syndrome, aspirin, salicylate, acetaminophen.*

When Ralph Douglas Kenneth Reye first described the syndrome that bears his name, he reported 21 cases of children admitted to the Royal Alexandra Hospital for Children in Sydney, Australia (now renamed The Children's Hospital, Camperdown) with a syndrome of encephalopathy and fatty degeneration of the viscera.<sup>1</sup> These 21 children had been seen at The Children's Hospital between 1951 and 1962 and ranged in age from 5

months to 8½ years. Until his death in 1977,<sup>2</sup> Reye continued to be actively involved in the pathologic diagnosis of the syndrome and the possible etiologic factors responsible for causing the syndrome. The cause of the syndrome eluded discovery by Reye as it has continued to elude us today.

In this report, we describe 26 children with Reye syndrome seen at the The Children's Hospital, Camperdown in Sydney, Australia, between 1973 and 1982.

### METHODS

The medical records of The Children's Hospital were searched for all cases of Reye syndrome diagnosed between 1973 and 1985. These cases were cross-checked with separate records kept in the department of pathology. The charts were reviewed and only cases fitting the Centers for Disease Control case definition of Reye syndrome were accepted. The Centers for Disease Control case definition is: (1) acute noninflammatory encephalopathy documented clinically by an alteration in consciousness, and, if available, a record of CSF containing eight leukocytes or fewer per microliter, or by histologic specimen demonstrating cerebral edema without perivascular or meningeal inflammation; (2) hepatopathy documented either by a liver biopsy or autopsy considered to be diagnostic of Reye syndrome or a threefold or greater increase in the levels of SGOT, SGPT, or serum ammonia; and (3) no more reasonable explanation for the cerebral or hepatic abnormalities.<sup>3</sup>

The case records were reviewed for medical histories, hospital course, laboratory studies, medication histories, toxicology screens, viral titers, and viral isolation studies. The medical records were detailed with respect to medication histories and drug levels, both because The Children's Hospital is a teaching hospital and because of Reye's interest in determining the causative agent for Reye syn-

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drome, a tradition that has been continued since his death.

Because of the reported association of Reye syndrome in the United States with the ingestion of aspirin and other salicylate-containing products,<sup>4-7</sup> the medications were carefully checked for composition in *Medical Intercontinental Medical Statistics*<sup>7a</sup> and a general usage review for aspirin and paracetamol (acetaminophen) as well as pediatric formulations of both products for the time period of the study was conducted.

## RESULTS

Of the 26 cases diagnosed at the Royal Alexandra Hospital for Children between 1973 and 1982, one case record was missing and 20 case records fit the Centers for Disease Control criteria for Reye syndrome (Table). Of these patients, only one child had ingested aspirin or salicylate-containing products, and this child had been vomiting repeatedly and had a zero salicylate level on admission to the hospital. None of the excluded five patients had ingested aspirin or salicylates. These excluded five were subsequently shown to have other diseases as explanations of their encephalopathy and/or hepatopathy. Two children had postvaricella cerebellar ataxia, and one child each had adipic aciduria,  $\beta$ -methyl glutaconic aciduria, and eucalyptus oil

ingestion. Six patients had ingested paracetamol (acetaminophen), seven patients had received ampicillin, three erythromycin, two trimethoprim-sulfamethoxazole, two a combination product containing dexchlorpheniramine maleate, phenylephrine, ammonium chloride, sodium citrate, and guaiphenesin, and the remaining medicines had been ingested by only one patient each and none contained salicylates. Seven patients had received no prescription or over-the-counter medicines prior to or during their illnesses. The patients ranged in age from 3 months to 7 years, with a median age of 22 months.

Compared with the US studies showing an association of Reye syndrome with aspirin or salicylate ingestion,<sup>4-7</sup> there was a high percentage of pathologic confirmation of the diagnosis in the Australian cases. Of 20 cases, 18 (90%) were in children who had liver biopsy or autopsy confirmation of the diagnosis, compared with less than 20% in the US studies.<sup>4-7</sup> The autopsied cases at The Children's Hospital showed the classic fatty degeneration of multiple organs (kidney, heart, and liver) and pancreatitis, as originally described by Reye. The pathologic specimens of four of the seven earliest cases in this series were personally reviewed by Reye.

Reye syndrome was first described in Sydney, Australia, and independently in the same year Johnson et al<sup>8</sup> described 16 cases of an encephalitis-

**TABLE.** Characteristics of Children With Reye Syndrome From the Royal Alexandra Hospital for Children in Sydney, Australia, 1973 to 1982\*

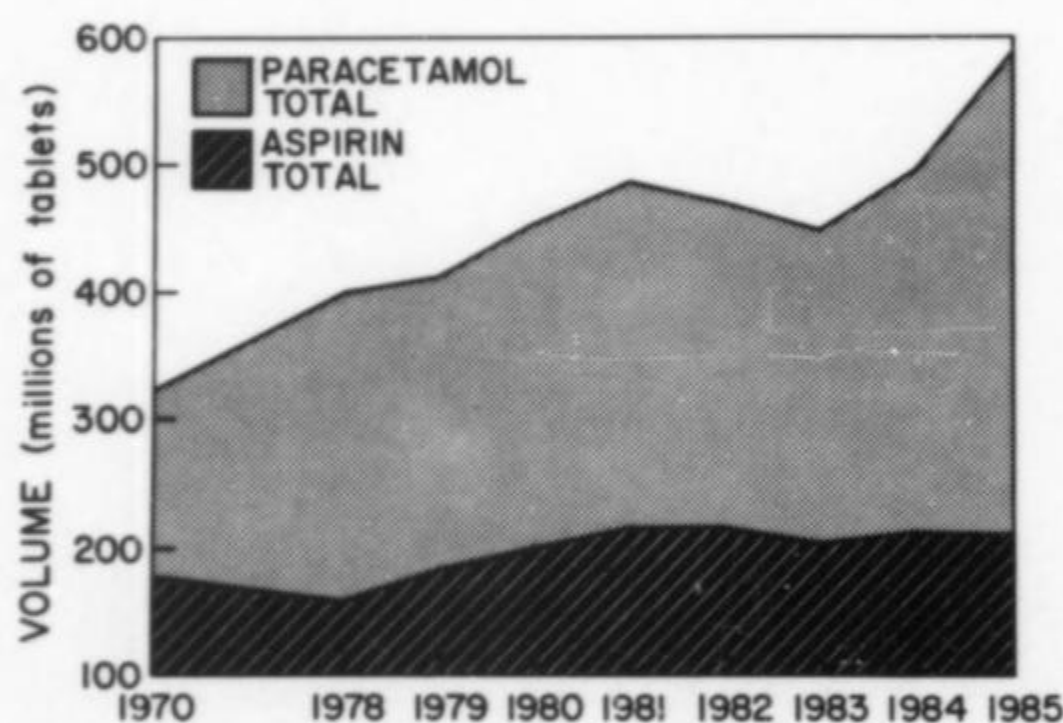
| Case No. | Age (yr) | Sex | Encephalopathy Criteria |                          |                 | Hepatopathy Criteria |                                                   | Medication Use |                             | Outcome  | Initial/Highest Stage of Illness <sup>23</sup> |
|----------|----------|-----|-------------------------|--------------------------|-----------------|----------------------|---------------------------------------------------|----------------|-----------------------------|----------|------------------------------------------------|
|          |          |     | Altered Consciousness   | CSF With WBC <8/ $\mu$ L | Brain Histology | Liver Biopsy         | >3 $\times$ Normal SGOT, SGPT, or NH <sub>3</sub> | Aspirin        | Paracetamol (Acetaminophen) |          |                                                |
| 1        | 11/12    | M   | +                       | +                        | +               | +                    | -                                                 | No             | Yes                         | Died     | V/V                                            |
| 2        | 5 1/12   | F   | +                       | -                        | +               | +                    | -                                                 | No             | No                          | Died     | V/V                                            |
| 3        | 7        | M   | +                       | +                        | +               | +                    | +                                                 | Yes            | No                          | Died     | IV/V                                           |
| 4        | 4        | F   | +                       | +                        | +               | +                    | +                                                 | No             | No                          | Died     | II/V                                           |
| 5        | 2        | M   | +                       | +                        | -               | +                    | +                                                 | No             | Yes                         | Good     | II/III                                         |
| 6        | 2 2/12   | F   | +                       | +                        | +               | +                    | +                                                 | No             | Yes                         | Died     | V/V                                            |
| 7        | 11/12    | M   | +                       | +                        | -               | -                    | +                                                 | No             | No                          | Good     | II/II                                          |
| 8        | 1 2/12   | F   | +                       | +                        | -               | +                    | -                                                 | No             | No                          | Good     | II/III                                         |
| 9        | 4 1/12   | M   | +                       | +                        | -               | +                    | -                                                 | No             | Yes                         | Good     | II/IV                                          |
| 10       | 5/12     | M   | +                       | +                        | +               | +                    | +                                                 | No             | No                          | Died     | II/V                                           |
| 11       | 9/12     | M   | +                       | +                        | +               | +                    | +                                                 | No             | Yes                         | Died     | IV/V                                           |
| 12       | 4/12     | M   | +                       | +                        | -               | +                    | +                                                 | No             | No                          | Survived | II/III                                         |
| 13       | 1 3/12   | F   | +                       | +                        | -               | -                    | +                                                 | No             | No                          | Good     | II/II                                          |
| 14       | 5/12     | M   | +                       | +                        | -               | +                    | +                                                 | No             | No                          | Good     | III/III                                        |
| 15       | 8/12     | M   | +                       | +                        | +               | +                    | +                                                 | No             | No                          | Died     | III/V                                          |
| 16       | 1        | F   | +                       | +                        | -               | +                    | +                                                 | No             | Yes                         | Good     | II/IV                                          |
| 17       | 5/12     | F   | +                       | +                        | -               | +                    | +                                                 | No             | No                          | Good     | II/III                                         |
| 18       | 3/12     | F   | +                       | -                        | -               | +                    | +                                                 | No             | No                          | Good     | II/IV                                          |
| 19       | 3        | F   | +                       | -                        | +               | +                    | -                                                 | No             | No                          | Died     | V/V                                            |
| 20       | 9/12     | M   | +                       | +                        | -               | +                    | +                                                 | No             | No                          | Survived | III/III                                        |

\* Symbols: +, present; -, absent.

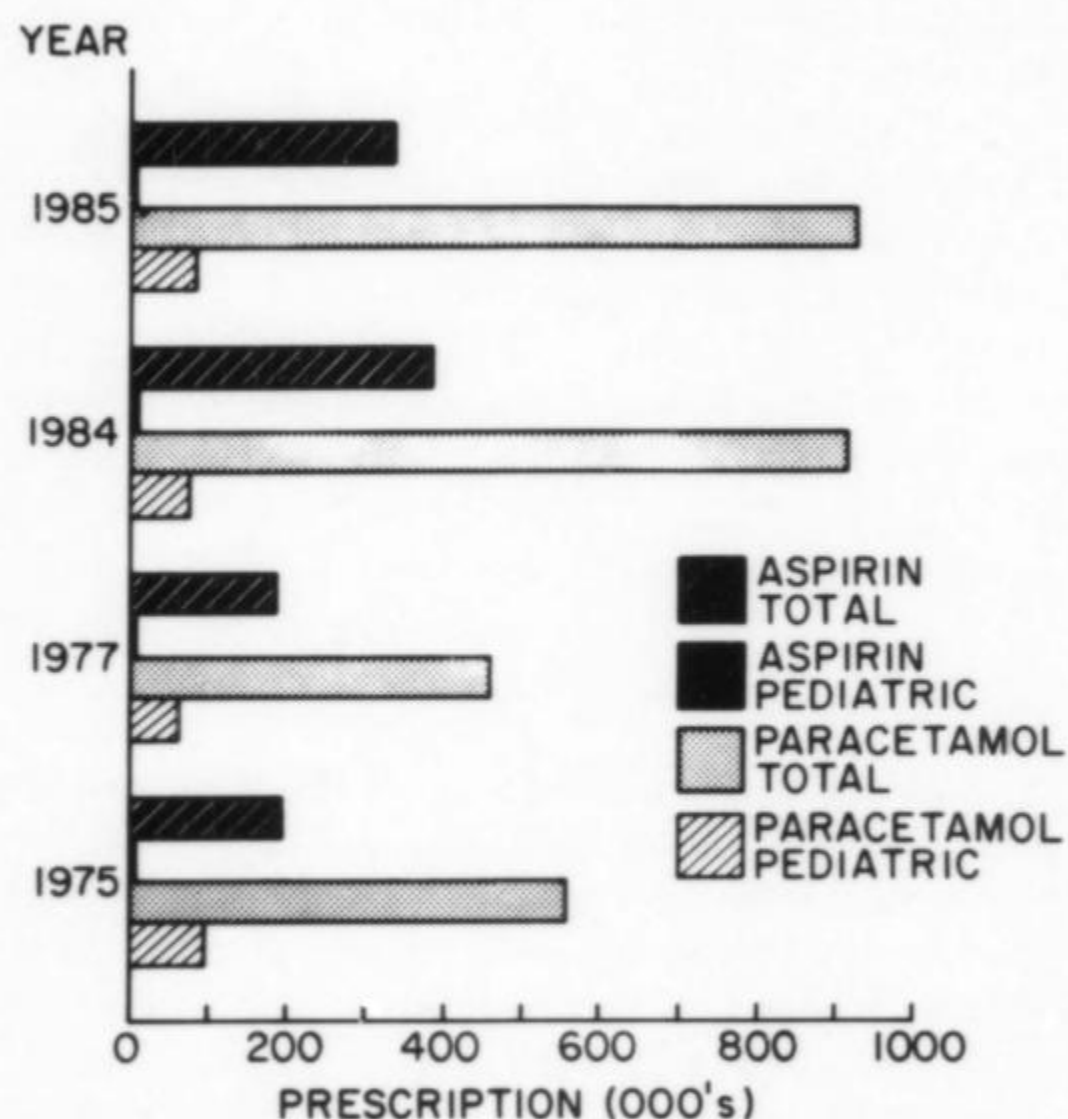


like illness in North Carolina, some of which were probably Reye syndrome. Although there is no central registry or organized reporting of Reye syndrome in Australia, data suggest that Reye syndrome is as common in Australia as in the United States. The CDC estimates the incidence of Reye syndrome in the United States to be ten to 20 cases per 1 million children younger than 18 years of age.<sup>9</sup> We found 20 cases of Reye syndrome at the Royal Alexandra Hospital for Children from 1973 to 1982, which compares with the 21 cases Reye reported between 1951 and 1962 in his original paper. If it is assumed that there are 1.5 million children (younger than 18 years) in the State of New South Wales (of which Sydney is the largest city) and that  $\frac{2}{3}$  of all hospitalized children with Reye syndrome in New South Wales would go to The Children's Hospital, the average incidence for stage II and greater Reye syndrome for the period of 1973 to 1982 would be 2.0 cases per 1 million children. Extrapolating these stage II and higher hospitalized patients to all children with Reye syndrome by the method of Lichtenstein et al<sup>10</sup> gives an incidence of nine cases per 1 million children. The 20 children had a mortality rate of 45% as compared with a 32% case-fatality rate in the United States during 10 years.<sup>3</sup>

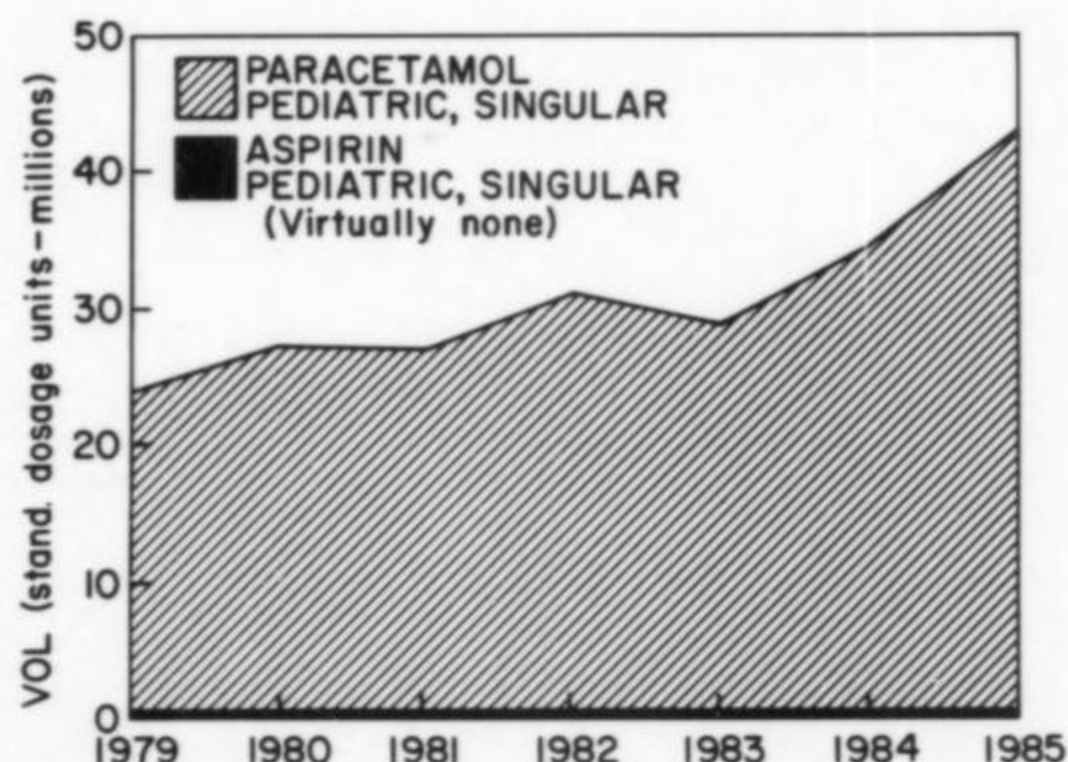
Paracetamol or acetaminophen has been the most commonly used analgesic/antipyretic in Australia for more than 25 years (Fig 1). This is true for both over-the-counter and prescription forms (Fig 2). For specific pediatric analgesic/antipyretic medications, aspirin usage has been virtually nil, with paracetamol dominating the pediatric analgesic/antipyretic market (Fig 3). For the period of this study, aspirin accounted for only 0.3% and paracetamol or acetaminophen for 99.7% of all sales



**Fig 1.** Analgesic sales through retail pharmacies from 1970 to 1985 expressed in millions of tablets. Paracetamol (acetaminophen) has experienced steady growth in sales, whereas sales of aspirin have remained relatively constant during this period.



**Fig 2.** Analgesic prescriptions for aspirin and paracetamol (acetaminophen); both total number of prescriptions and prescriptions for pediatric preparations from 1975 through 1985. Pediatric aspirin prescriptions have been negligible.



**Fig 3.** Sales through retail pharmacies of pediatric forms of analgesics, paracetamol (acetaminophen) and aspirin, expressed in standard dosage units during 1979 through 1985. There have been virtually no sales of pediatric aspirin preparations during this period and paracetamol has clearly dominated the pediatric analgesic market.

of pediatric analgesic/antipyretic products based on dosage units.

One patient with autopsy-confirmed Reye syndrome had a salicylate level of 0.2 mg/100 mL despite a negative history for salicylate ingestion specifically stated in the chart. This was presumed to be a false-positive finding.<sup>11</sup>

Despite viral cultures and titers having been performed on many of the patients, there was no clear-



cut association of any particular virus with Reye syndrome at The Children's Hospital. Varicella, respiratory syncytial virus, and coxsackie B were documented in one case each, and parainfluenza type 1 virus was isolated from two cases. In four cases, the patient had received a vaccination (diphtheria-pertussis-tetanus in three, measles in one) within 1 to 2 weeks of the onset of the Reye syndrome.

## DISCUSSION

Results of three retrospective, case-control public health department studies<sup>4-6</sup> and a US Public Health Service pilot study<sup>7</sup> have all suggested an association between the ingestion of aspirin or salicylates and the development of Reye syndrome. Although the cause of Reye syndrome remains a mystery, on the basis of these statistically significant associations, the government launched a massive public education campaign in late 1982 to caution parents, children, and teenagers to avoid salicylates during respiratory illnesses or chicken pox. Millions of dollars have been expended on an illness that is responsible for one or two deaths per year per 1 million children younger than 18 years of age. Articles have now begun to appear in which it is suggested that a decrease in the incidence of Reye syndrome is commensurate with a decrease in the use of aspirin.<sup>12</sup>

As impressive as the statistically significant associations appear on the surface, they seem to contradict logic. It is unlikely that a drug that has been used for more than 100 years could be responsible for a syndrome that was first described only 25 years ago. Scientists have been unable to produce a Reye-like illness in animals using aspirin, although they have been able to do so with acetaminophen and other toxins.<sup>13,14</sup> The only study to produce hepatic lesions similar to those in Reye syndrome with aspirin required toxic levels of salicylate.<sup>15</sup> The experts agree that Reye syndrome is not due to toxic levels of aspirin, and the studies reporting an association have all cautioned that the amounts of salicylates ingested have been reasonable to low. The only animal model of Reye syndrome demonstrating a synergism between influenza infection and aspirin treatment requires the animals to be hyperammonemic and comatose from an arginine-deficient diet prior to infection and treatment.<sup>16</sup>

We now report a retrospective chart-review study from The Children's Hospital, Camperdown, Sydney, Australia, where Reye syndrome was first described and where Reye syndrome is as common as in the United States, showing a complete lack of

association between the development of Reye syndrome and the ingestion of aspirin or salicylates. This lack of association is not unexpected, because aspirin use in Australia, especially in the pediatric age group, has been practically nonexistent since before the time Reye first described the syndrome. Paracetamol or acetaminophen has been the most frequently used analgesic and antipyretic for at least 25 years. Despite this lack of association of Reye syndrome with aspirin use, Reye syndrome has been as common in Australia as in the United States. A purposeful underestimate of the incidence of Reye syndrome in Australia was obtained by dividing the number of cases of Reye syndrome seen at a single hospital (The Children's Hospital, Camperdown) in Sydney, New South Wales, by the total pediatric population younger than 18 years of age in New South Wales, and by assuming that  $\frac{2}{3}$  of all patients with Reye syndrome went to The Children's Hospital. This is an obvious underestimate of the incidence of Reye syndrome in New South Wales because there is one other pediatric intensive care unit and numerous other hospitals serving the pediatric population of New South Wales. In addition, all patients with Reye syndrome admitted to The Children's Hospital were in stage II or greater. We then extrapolated the incidence of Reye syndrome in the general population from the hospitalized patients using published data.<sup>9,10</sup>

It is difficult to reconcile this lack of association between aspirin use and Reye syndrome in Australia with the strong and statistically significant associations reported by the United States and individual state public health services.<sup>4-7</sup> A published study from Japan has also failed to show an association between aspirin use and Reye syndrome.<sup>17</sup> Perhaps, the association of aspirin use with the development of Reye syndrome in the United States is purely coincidental.

The US public health studies are not without fault. They were all retrospective, although this study is also. It is possible that surreptitious aspirin use was missed in this chart review study, but the data in this study fit well with the nationwide data on the use of salicylates in Australia, especially in children. The medical records of the patients with Reye syndrome were thorough in their medication-use histories, partly because The Children's Hospital is a teaching hospital and partly because R.D.K. Reye was from The Children's Hospital and was interested in unraveling the etiology of the syndrome that bears his name. Reye (personal communication, March 1977) had even personally entertained the possibility that aspirin might play a role because one of his original cases was in a child with juvenile rheumatoid arthritis who was on



chronic salicylate therapy. Further analysis of medication histories convinced him that salicylates were not responsible for the syndrome.

One of the criticisms of the public health studies has been the lack of pathologic confirmation of the diagnosis of Reye syndrome (<20% in all the studies).<sup>18</sup> Although a clinical diagnosis of Reye syndrome is acceptable for initiating therapy and clinically managing a case of Reye syndrome, pathologic confirmation of the diagnosis should be required for scientific studies.<sup>10,19-21</sup> Reye syndrome has distinct and specific pathologic findings<sup>1,10,19,21</sup> and a number of clinical syndromes can mimic Reye syndrome.<sup>18,21,22</sup> As opposed to the low percentage of pathologic confirmation of the diagnosis of Reye syndrome in the studies purporting an association of aspirin use with the development of Reye syndrome, this study had a 90% pathologic confirmation of the diagnosis and no association with aspirin use. The similarities between the Australian data presented here and data from Great Britain are notable. The annual incidence of Reye syndrome in Great Britain has ranged from three to seven cases per 1 million children, with a median age of 14 to 15 months, and the overall mortality has been 50%.<sup>23</sup>

It is not unexpected that studies would begin to report an associated decline in the incidence of Reye syndrome in the United States commensurate with a decline in aspirin use.<sup>12</sup> Critical review of the national data suggests a decline in the incidence of Reye syndrome beginning before the public awareness campaigns.<sup>3</sup> It is interesting to note that the incidence of Reye syndrome appears to be declining in Australia also, despite lack of association of Reye syndrome with aspirin use. There is no reason to believe that Reye syndrome is a different disease in the United States than in Australia. The cause or causes of Reye syndrome remain a mystery. It is unlikely that the purported association of Reye syndrome with aspirin use is anything more than coincidental.

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